What MORDRED can tell us about

12 popular narratives

concerning the economy and the environment

Three pillars project revolutio-N e.V.

1. "The global economy does not cause environmental problems."

False. The way the global economy works today directly leads to environmental destruction.

In the simulations, producing and consuming more and more inevitably leads to higher emissions, more mining, more land use changes, and more pollution. These problems are not accidents or side effects—they are built into how the global economy currently operates.

2. "Environmental problems won't really affect economic growth, human development or quality of life."

False. Environmental damage can stop growth and make life worse for billions of people.

It is highly plausible that climate change and ecological damage will affect the economy, slowing, stopping or even reversing growth. As the capacity of the economy to produce goods shrinks, and more labor is needed to temporarily "fix" environmental problems, per capita consumption, quality of life and life expectancy start to fall. Environmental damage doesn't stay "outside" the economy—it comes back and hits people directly.

3. "Environmental problems will hurt only poor people in poor countries."

False. The poor in the Global South are hit first and hardest—but everyone is eventually affected.

The simulations show that poorer regions suffer earlier: more deaths, slower development, and deeper crises. But rich countries and rich people are not protected forever. Because the world economy is deeply connected, shocks spread, and living standards in rich countries can also fall sharply. 'Leaving the poor to die' can cause the global economy to collapse and hurt the rich, since their consumption is made possible through the cheap labor of the poor.

4. "Technology will fix environmental problems while everyone keeps getting richer."

False. Technological change helps and is necessary, but it is not enough to solve the problem on its own.

Cleaner technologies reduce some damage, but if total production and consumption keep growing, the overall pressure on the planet keeps increasing. Problems don't disappear—they shift from climate to land, water, materials, or pollution. Furthermore, technological changes take time – from years to decades – during which environmental deterioration continues. The idea that everyone can endlessly get richer due to technological fixes does not hold in the simulations.

5. "Switching from fossil fuels to renewable electricity will solve the problem."

False. 'Green' electricity is necessary—but far from sufficient.

Even when electricity becomes almost fully renewable, many problems remain: high non-electric fossil energy use, material extraction, land conflicts, pollution, and climate damages that are already unavoidable. 'Fixing' electricity is only fixing one part of a much bigger problem

6. "Fair distribution has nothing to do with environmental problems."

False. Inequality is a central part of the problem.

Who consumes how much matters enormously. A small group with very high consumption creates a large share of environmental damage. The simulations show that egalitarian societies that consume at low levels are better able to reduce pressure on the planet and protect human well-being at the same time.

7. “Reducing the consumption of the rich is unnecessary and harmful.”

False. Reducing excessive consumption at the top is both necessary and beneficial.

When the richest people, countries and regions reduce consumption and resources are shared more fairly, environmental damage falls and living conditions improve for the majority. High consumption by the rich does not only require a lot of materials and energy. It also shapes social norms and development goals, driving rising material expectations everywhere and making environmental breakdown arrive faster and more violently.

8. “Rich countries are the key actors in solving the problem.”

False. Rich countries of the Global North matter, but due to their population size and resources they won’t be the main drivers of future impacts.

As many countries in the Global South grow and industrialize, their development paths will strongly shape global outcomes. The simulations show that global solutions imply redistribution, sufficiency and technological change everywhere - not just in rich countries. Big emerging countries will be decisive in transforming economic production and consumption processes, and shaping a world order with just rules governing the world economy.

9. “Population size doesn’t matter.”

False. Population matters – but it’s complicated.

Simulations in which world population peaks early and declines strongly are linked to a significant reduction of economic pressure on the planet. Population, consumption, and technology interact—and ignoring any of them makes it impossible to address environmental problems.

10. “It doesn’t matter whether we act now or later.”

False. Delay makes everything worse.

Early action increases the chances of successful societal change and avoids major damage on societies and ecosystems. Late action happens after ecosystems are damaged and resources are depleted, leading to sharper declines, more suffering, higher risks and fewer options.

11. “The future is urban: subsistence and peasant economies will disappear.”

False. Subsistence economies do not disappear automatically; their future depends on economic stability, land access, and crisis dynamics.

In a future without environmental harm affecting the global economy, people voluntarily or involuntarily move into cities and subsistence declines. In others, economic collapse and climate stress push people back to rural survival strategies. Subsistence is not just a “leftover from the past”—it could become a last line of survival when the global economy fails.

12. “If every country invests in green technology, everything will be fine.”

False. This is overly optimistic and misleading.

Simulations show that national green technology strategies alone are insufficient. The only scenarios that approach long-term environmental stability achieve higher recycling rates, re-use of materials, and a focus on the production of basic economic goods through strong global coordination, rapid global redistribution and shared rules for resource use. This means that what works best is a world that acts together: sharing resources, strongly limiting excessive consumption, and coordinating transitions globally. Without this kind of planetary cooperation, green technology slows the crisis —but does not stop it.

More Information

MORDRED

MORDRED is a computer model that allows to simulate different scenarios of future changes in the global economic, demographic, land, energy and climate system. While it cannot predict the future of the world it can be used to assess to which extent different powerful ideas influencing current political decision-making are true or false.

SIMULATIONS

Simulations with MORDRED are not perfect. But they can give us a very broad idea of what to expect from the future. Without strong growth in recycling and material reuse rates and a scaling up of renewable energy technologies, a reduction of population growth and the elimination of harmful and superfluous consumption goods Earth system parameters will continue to deteriorate, creating deep and high risks for human well-being. Under the current world order, societies lack the capacities to change their ways of production and consumption at the required rates and scales. For societies to acquire these capacities, dominant political, economic and cultural ideas and rules of the game must be changed at the global level. In other words: a good life for all during the 21st century and beyond requires striving for a new world order.

BOOK

If you are interested in detailed and technical information on the MORDRED model, the scenarios simulated, and the meaning of the scenario results, download the book ‘Putting numbers to ideas. Using MORDRED to explore dreams and nightmares about the future of the world’.